Foundations of Machine Learning

Quiz 2 - Notes

Regression :-

Regression problems

- → Nature of relationship
  - → Linear
  - → Non-Linear
- → No. of dependent and independent variables
  - → Univariate (simple regression) → One dependent & one independent
  - → Multivariate regression → Multiple dependent and multiple independent variables
  - → Multiple Regression → Single dependent and Multiple independent variables
- → Types of ~~Multiple~~ Solution approaches based on errors
  - → Ordinary Least Squares (OLS)
  - → Total Least Squares (TLS)

Note:- Relation b/w dependent & independent variables is assumed to be a linear function of the parameters (regression coefficients). Function need not be linear in the variables.

True value :- y^t (unknown)

Observed value :- y

Predicted value :- $\hat{y}$

Simple Linear Regression :- (SD - SI)

$$\boxed{y(x, w) = w_0 + w_1 x + \varepsilon} \Rightarrow y = y^t + \varepsilon$$

Multi Linear Regression :- (SD-MI)

$$\boxed{y(x, w) = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_D x_D + \varepsilon}$$

Generalization :-

$$\boxed{y(x, w) = w_0 + w_1 \phi_1(x) + w_2 \phi_2(x) + \cdots + w_D \phi_D(x) + \varepsilon}$$

Multivariate Linear Regression :- (MD-MI)

$$y_1(x, w) = w_{01} + w_{11} x_1 + w_{21} x_2 + \cdots + w_{D1} x_D + \varepsilon_1$$

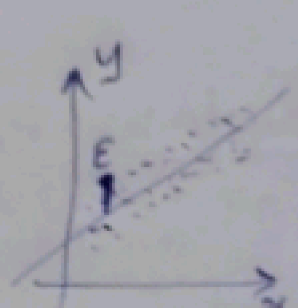
$$y_2(x, w) = w_{02} + w_{12} x_1 + w_{22} x_2 + \cdots + w_{D2} x_D + \varepsilon_2$$

$$\vdots$$

$$y_m(x, w) = w_{0m} + w_{1m} x_1 + w_{2m} x_2 + \cdots + w_{Dm} x_D + \varepsilon_m$$

Simple and Multiple Linear Regression :-

$$\boxed{y = [1_n \; X] w + \varepsilon} \Rightarrow \boxed{y = X^* w + \varepsilon}$$



Assumptions on Random Error (ε):-

* $\boxed{E[\varepsilon] = 0}$ $\rightsquigarrow$ actual values of the output should be equally spread around than predicted value.

* **Homoscedastic Errors:-**

$\boxed{Cov[\varepsilon] = \sigma^2 I}$ $\rightsquigarrow$ Variance of error term should be same for all values of independent variables and errors across different observations should not be correlated.

* Errors should be normally distributed.

* Independent variables and error terms should not be correlated.

Ordinary Least Squares (OLS):- (closed form Solution)

Assumptions:-

* No Multi-collinearity (Independent variables should not be highly correlated
* No. of samples in the dataset should be greater than the no. of regression coefficients.
* 'X' is observed without noise; noise is in 'y' alone.

Residual:- $\boxed{r = y - \hat{y}}$; $\boxed{\hat{y} = X^* w + \varepsilon}$

$$\boxed{SSR = (y - X^* w)^T (y - X^* w)}$$

Least Squares estimate of w are the coefficients that minimize SSR.

$$\boxed{\hat{w} = \underset{w}{\arg\min}\ SSR} \rightsquigarrow \boxed{\hat{w} = (X^{*T} X^*)^{-1} X^{*T} y}$$

OLS Solution = MLE :-

$$\boxed{\text{Max. } \log P(y|x,w) = \text{Min. } \frac{1}{2\sigma^2}\sum_{i=1}^{N}(y_i - x_i^T w)^2}$$

(Gaussian distribution of noise)

Interpretation of $\hat{\omega}$:-

* Best fit under the assumptions made
* It implies, $\hat{\omega}$ is unbiased estimate of w.
* If assumptions fail, $\hat{\omega}$ would be biased.
* w_i - change in the dependent variable for a unit change in the corresponding independent variable x_i, holding everything constant.

LA interpretation of OLS :-

→ variables (or features) as dimensions

→ Samples as dimensions

→ OLS in matrix form :-

$$\boxed{\hat{y} = X^* \hat{\omega}}$$

→ $\hat{y}$ lies in the column space of X^*

→ $\hat{y}$ is the projection of y onto the column space of X^*.

→ Projection matrix :-

$$\hat{y} = X^* \hat{\omega}$$

$$\hat{y} = X^* (X^{*T} X^*)^{-1} X^{*T} y$$

$$\boxed{\hat{y} = Py}$$

→ Residual vector is ⊥r to the column space of X^*

$$r = y - X^* \hat{\omega}$$

Proof :- $X^{*T} . r = 0$

Total Least Squares (TLS):- (No closed form solution)

Note:- Right Singular Vector + Largest Singular Value
= direction of maximum variance in the data.

$$X = USV^T$$

→ This can be written as the sum of r-Rank 1 matrices.

Frobenius Norm:-

$$\|X\|_F = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} X_{ij}^2}$$

* Assumptions:- The error is both in dependent and independent variables.

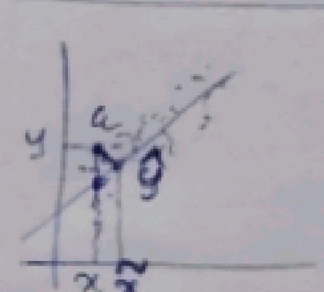
True model:- $y^t = w_0 + w_1 x^t$

observed model:- $y = w_0 + w_1 x^t + (w_1 \varepsilon_x - \varepsilon_y)$

Objective function of TLS:- $\min_{w} \|[x, y] - [\hat{x}, \hat{y}]\|_F$

s.t. $\hat{y} \in \text{colspace}(\hat{x})$

$\hat{y} = \hat{x}\hat{w}$



TLS Solution Steps:-

Step 1:- Scale the data pertaining to all independent and dependent variables to zero mean and unit variance.

Step 2:- Construct the augmented data matrix combining the input and output data matrices

$$X_{aug} = [X \ y]_{N*(D+1)}$$

Step 3:- Perform SVD on the augmented data matrix

$$X_{aug} = USV^T$$

Step 4:- Identify the smallest singular value and its corresponding right singular vector.

→ σ_{D+1} is the smallest singular value with right singular vector V_{D+1}

Step 5:- Partition the singular vector V_{D+1} corresponding to independent and dependent ~~vector~~ variables as follows.

$$V_{D+1} \in \mathbb{R}^{D+1} \ ; \ V_{D+1} = \begin{bmatrix} V_x \\ V_y \end{bmatrix} \ ; \ \begin{matrix} V_x \rightarrow D \times 1 \\ V_y \rightarrow 1 \times 1 \end{matrix}$$

Step 6:- Estimates for regression coefficients are computed as follows:-

$$\boxed{\hat{\omega} = \frac{-1}{V_y}[V_x]}$$

$$y = X\omega$$

$$[X \ y]\begin{bmatrix} \omega \\ -1 \end{bmatrix} = 0$$

$$\boxed{X_{aug} \, v = 0} \implies$$ v should lie in the null space of X_{aug},

but it is not easy as there is no closed form solution, it is derived using the $\hat{\omega} = \frac{-1}{V_y}[V_x]$ formula.

→ V_{D+1} is the direction ⊥r to the direction of maximum variance in the data.

Step 7:- Get corrected estimates for the data matrices X and y by taking the low rank matrix approximation of the augmented data matrix.

$$X^{c}_{aug} = u_1 \sigma_1 v_1^T + u_2 \sigma_2 v_2^T + \cdots\cdots + u_D \sigma_D v_D^T$$

Special cases of OLS & TLS:-

→ Error variance differs across observations::

$$var(\varepsilon_i) = \sigma_i^2$$

→ This leads to heteroscedasticity.

→ Standard OLS/TLS are no more estimators with least variance among unbiased estimators.

→ Weighted OLS/TLS resolves this by factoring the heteroscedasticity in noise while estimating the parameters

→ weighted OLS solution:-

$$\boxed{\hat{\omega} = (X^{*T} \Sigma^{-1} X^{*})^{-1} X^{*T} \Sigma^{-1} y}$$

$$\Sigma = diag(\sigma_1^2, \sigma_2^2, \ldots\ldots, \sigma_N^2)$$

→ Approach 1:- Reliability based weights

Approach 2:- Iterative re-weighting

→ In weighted TLS, only the step 2 & 3 of TLS gets modified,

Step 2 :- $X^{S}_{aug} = \Sigma^{1/2} [X \ y]_{N*(D+1)}$

Step 3 :- Perform SVD on the new augmented matrix

$$X^{S}_{aug} = USV^{T}$$

Follow the remaining steps as it is in TLS.

Polynomial Regression :-

$$y = w_0 + w_1 x_1 + w_2 x_1^2 + w_3 x_2 + w_4 x_2^2 + w_5 x_1 x_2$$

→ OLS can be used to find these coefficients

→ Overfitting can be an issue because many terms get added with every additional ~~term~~ degree of polynomial

→ Degree of the polynomial can be decided using cross-validation with regression metrics.

Logarithmic Regression :-

→ Use when the variable has a skewed distribution

→ Use when known exponential relation, $y = ae^{bx}$

→ Use when the variance in error is not constant.

$\ln(y) = \ln(a) + bx$

$y_1 = w_0 + w_1 x$

→ Use when there is a wide range in the distribution of the data.

→ Use when there are percentage changes (relative changes) and not absolute changes. $\frac{d(\ln y)}{dy} = \frac{1}{y}$

$\Rightarrow d(\ln y) = \frac{dy}{y} \Rightarrow d(\ln y) = y_1 - y_2 / y_1 \Rightarrow$ % change.

Log-Linear model:-

$$\boxed{\ln y = w_0 + w_1 x}$$

Linear-Log model:-

$$\boxed{y = w_0 + w_1 \ln x}$$

→ OLS can be used to find the regression coefficients.

→ Log transformation can be applied only for variables taking positive values. Shifting can be done in case of 0 values.

→ Caution:- If range of x is small, then there could be a high correlation b/w x and $\ln x$ | x and x^2.

Multi-variate Linear Regression:-

Assumptions:-

* Errors are independent across samples of an output but error covariance across outputs is allowed.

* Errors for each output, across samples are jointly normal (multi-variate normal).

Two solution approach:-

→ MISO

→ MIMO

Limitations of Closed form solution:-

* Computing $(X^{*T} X^*)^{-1}$ is costly
* Expensive for large datasets
* Requires storing the entire data in memory.
* Cannot be used when data is streaming in packets/batches.

Gradient Descent and Loss function:- (for simple linear regression)

$$J(W) = \frac{1}{2N}(y - X^* w)^T (y - X^* w)$$

$$\boxed{J(w) = \frac{1}{2N} \sum_{i=1}^{N} (y_i - x_i^T w)^2}$$

Properties of loss functions:-

* Quadratic and convex
* Global minimum exists
* Ideal for gradient based optimization

Gradient Descent Approach:-

$$\boxed{w^{k+1} = w^k - \alpha \nabla J(w^k)}$$

$$\Rightarrow \nabla J(w) = \frac{-1}{N} X^T (y - Xw)$$

$$\Rightarrow w^{k+1} = w^k + \alpha \frac{1}{N} X^T (y - Xw)$$

Feature Scaling:-

$$x' = \frac{x - \bar{x}}{\sigma_x}$$

* Z-score standardization
* Min-max scaling

Bias-Variance Trade-off:-

Bias:- Bias in predictions measures how far average prediction is from true value.

$$\boxed{\text{Bias} = E[\hat{y}] - y^t}$$

Variance:- Variance measures change in model's predictions when trained on different datasets.

$$\boxed{\text{Var} = E[(\hat{y} - E[\hat{y}])^2]}$$

$$\boxed{MSE = Bias^2 + Variance + \text{Irreducible error}}$$

$$E[(y-\hat{y})^2] = (E[\hat{y}] - y^t)^2 + E[(\hat{y} - E[\hat{y}])^2] + \sigma_\varepsilon^2$$

* Increasing data reduces model variance but cannot remove bias.

Regularization:-

$$\boxed{\min_w (y - X^* w)^T (y - X^* w) + \lambda \,\Omega(w)}$$

λ - regularization rate ; $\Omega(w)$ - regularization penalty
$(\lambda \geq 0)$ $\Omega(w)$ is large for complex models and small for simpler models.

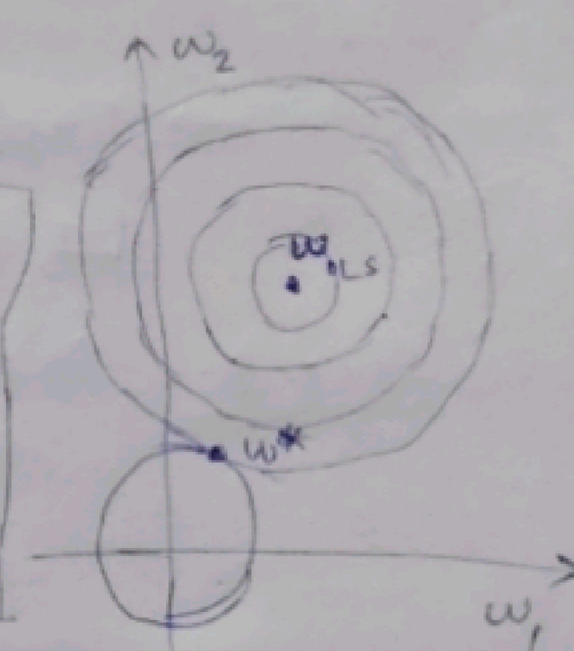
Impact of regularization:-

* Parameters are pushed towards zero
* Significant decrease in variance with small increase in bias

Ridge (L2) Regularization:-

$$\boxed{\min_w (y - X^* w)^T (y - X^* w) + \lambda \|w\|_2^2}$$

$$\boxed{w_{ridge} = (X^{*T} X^* + \lambda I)^{-1} X^{*T} y}$$

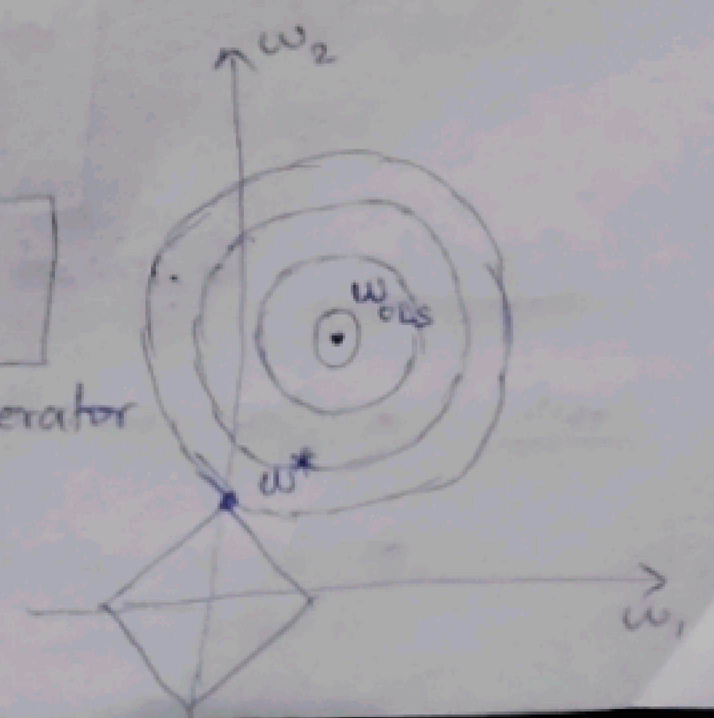


→ Penalizes squared magnitude of coefficients
→ Shrinks all coefficients smoothly with increase in λ.
→ Shrinks larger coefficients more.

Lasso (L1) Regularization:-

$$\boxed{\min_w (y - X^* w)^T (y - X^* w) + \lambda \|w\|_1}$$

→ Least absolute shrinkage and selection operator
→ Can drive some coefficients exactly to zero
→ Can be used for feature selection



Lasso using Coordinate Descent :-

Step 1 :- For each of the D predictors (inputs), compute partial residual for all samples.

$$r_{i(-j)} = y_i - \sum_{k \neq j} x_{ik} w_k \quad \forall j \text{ vars}$$

→ Gives what part of y_i is left unexplained if we ignore variable j

→ This is the target, w_j must explain.

Step 2 :- Compute OLS like contribution of feature j

→ Regress this only on x_j (all samples)

$$r_{-j} = e_j x_j$$

$$e_j = \frac{x_j^T r_{-j}}{x_j^T x_j} = x_j^T r_{-j} \quad \text{(scaled features)}$$

→ Explains potential value ~~for~~ w_j when there is no L1 penalty.

Step 3 :- Apply soft thresholding and update the weights

$$\boxed{w_j = \frac{e_j - \lambda}{\|x_j\|^2}} \; ; \; e_j > \lambda$$

$$\boxed{w_j = 0} \; ; \; |e_j| \leq \lambda$$

$$\boxed{w_j = \frac{e_j + \lambda}{\|x_j\|^2}} \; ; \; \cancel{e_j} \; e_j < -\lambda$$

Step 4 :- Repeat all 3 steps until convergence

Note :- * No gradients, no matrix inverse and natural sparsity

* λ is chosen using validation techniques

Model Validation :-

→ (Training dataset ; Validation dataset; Testing dataset)

* Hold out · (70:30 / 80:20 split of data)
* K-fold cross validation (K folds of N samples each with $\frac{N}{K}$ samples)
* Leave one out cross validation. (1 datapoint of N samples is used for cross validation)

Metrics for Regression model evaluation:-

* MAE (Mean Absolute Error) - Treats all errors equally (doesn't amplify)

$$\boxed{MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|}$$

* RMSE (Root Mean Squared Error) - Penalizes large errors more

$$\boxed{RMSE = \sqrt{\left(\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2\right)}}$$

* MAPE (Mean Absolute Percentage Error) - Useful to understand relative errors

$$\boxed{MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100}$$

Metrics to evaluate Goodness of fit:-

* R-squared (R^2) :- (Indicates how good is the fit)

$$\boxed{R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}}$$

→ +1 = overfitting ; -1 = worst fit ; 0 = same as mean.

Note:- To be used only to understand variance explained and compare models built on same dataset

Issues:- * can be misleading in case of low variance in dependent variable

* R^2 increases with additional variables even if irrelevant

* High R^2 does not imply good predictions (model may still have large errors).

* Adjusted R^2:- (To overcome the limitations of R^2)

$$\boxed{\text{Adj. } R^2 = 1 - \left(\frac{(1-R^2)(n-1)}{n-p-1}\right)}$$

Feature Selection:-

Objective:- Select a subset of features that ~~generalizes~~ minimizes generalization or validation error.

$$S \subset \{1, 2, \dots, D\}$$

$$\mathcal{L}_{val}(S) = \frac{1}{N_{val}} \sum_i (y_i - \hat{y}_i^{(S)})^2$$

→ Recursive Feature Elimination:-

Small $|w_j|$ → weak contribution → Candidate for removal

Algorithm:-

Let $S_0 = \{1, 2, \dots, D\}$. For each iteration $i = 0, 1, 2, 3 \dots$, Fit linear model using all features in S_i. Rank the features based on their coefficients $r_j = |w_j^{(i)}|$. Remove feature with smallest rank. Repeat until desired no. of features remain.

→ Sequential Feature Selection:-

Algorithm:- Start with an empty set $S_0 = \Phi$. At step i, for each candidate feature j not in S_i. Fit model using $S_i \cup \{j\}$ compute validation loss $\mathcal{L}_{val}(S_i \cup \{j\})$. Choose feature: $j^* = \arg\min_j \mathcal{L}_{val}(S_i \cup \{j\})$. Update the feature set: $S_{i+1} = S_i \cup \{j^*\}$.

Note:- This can be done backwards starting with complete set and eliminating one by one.

Bayesian Linear Regression:-

Estimate a distribution of parameters instead of point estimates. Assuming a Gaussian noise: $\varepsilon \sim \mathcal{N}(0, \beta^{-1})$.
Implies target values follow a Gaussian distribution,

$p(y|w) = \mathcal{N}(Xw, \beta^{-1}I)$. ; β - Precision (or) Inverse variance.

Assumes a prior distribution on parameters and infers a posterior based on data using Baye's rule.

$$\boxed{p(w|D) = \frac{p(D|w)\, p(w)}{p(D)}}$$

Non-Linear Regression:-

Eg:- $\ln y = A - \frac{B}{x+C}$: Antoine equation.

Objective function:- Minimize the sum of residuals between observed and predicted values.

$$\min_{A,B,C} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \qquad \text{s.t. } \ln \hat{y}_i = A - \frac{B}{x_i + C}$$